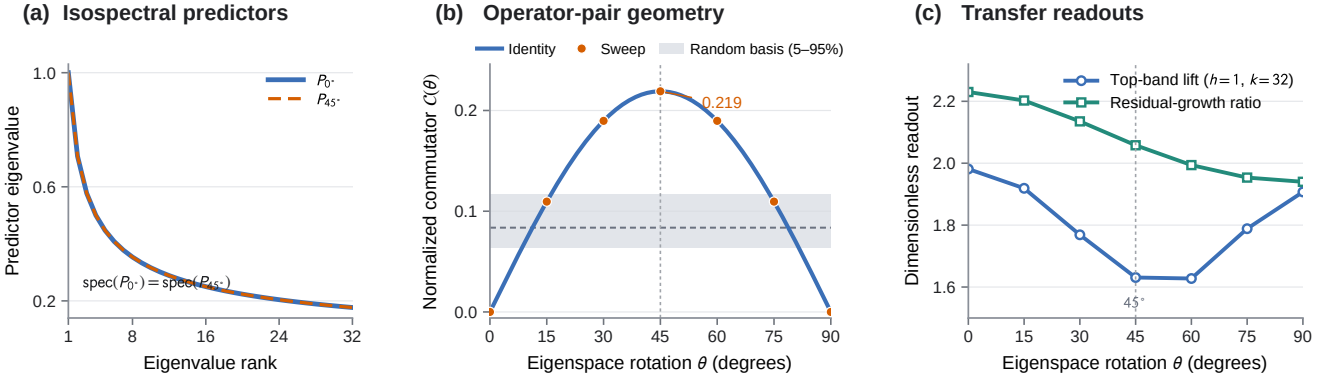
(a) Isospectral predictors
Predictor eigenvalue
1.0
0.6
0.2
P_{0°
P_{45°
spec(P_{0°) = spec(P_{45°)
1
8
16
24
32
Eigenvalue rank
(b) Operator-pair geometry
Identity
Sweep
Random basis (5–95%)
Normalized commutator $C(\theta)$
0.2
0.1
0.0
0.219
0
15
30
45
60
75
90
Eigenspace rotation θ (degrees)
(c) Transfer readouts
Dimensionless readout
2.2
2.0
1.8
1.6
Top-band lift ($h=1$, $k=32$)
Residual-growth ratio
45°
0
15
30
45
60
75
90
Eigenspace rotation θ (degrees)